VELS
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Neurocalc: AI-Powered Mathematical Problem-Solving Assistant

FINAL REVIEW PRESENTATION

REVIEW-03

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BATCH N0: 12

INTRODUCTION

- Traditional calculators are primarily designed to perform numerical computations and provide limited support for conceptual understanding or interactive problem-solving.
- Although Large Language Models (LLMs) enable natural language interaction, they may produce inaccurate or inconsistent results when solving complex mathematical problems.
- Students and educators require an intelligent system that combines accurate symbolic computation with clear, human-like explanations to enhance the learning experience.
- NeuroCalc is a hybrid AI-powered mathematical assistant that integrates a locally hosted Large Language Model (Ollama) with the SymPy symbolic mathematics library to deliver reliable solutions and contextual explanations.
- The proposed system provides an offline-oriented, privacy-focused, and user-friendly platform that supports mathematical learning through natural language interaction and deterministic computation.

ABSTRACT

NeuroCalc is an AI-powered mathematical problem-solving assistant that integrates a locally hosted Large Language Model (LLM) using Ollama with the SymPy symbolic mathematics library. It provides accurate mathematical computation and intelligent conceptual assistance through a user-friendly CustomTkinter graphical interface. A query routing module analyzes user input and directs mathematical expressions to SymPy for deterministic symbolic computation, while conceptual queries are processed by the LLM for natural-language explanations. This hybrid architecture combines computational accuracy with conversational intelligence. NeuroCalc is designed as an offline-oriented, privacy-focused educational tool, supporting students and educators with reliable solutions and clear explanations while reducing dependency on external cloud services.

Problem Statement:

- Traditional calculators provide accurate mathematical computations but lack the ability to explain concepts, reasoning, or solution steps.
- LLM-based systems support natural-language interaction but may generate inaccurate or inconsistent results for complex mathematical computations.
- Existing solutions often treat mathematical computation and intelligent explanation as separate functionalities, limiting their effectiveness as learning tools.
- Students and educators require a reliable and interactive system that can provide accurate mathematical solutions along with clear, understandable explanations.
- Therefore, there is a need for a unified solution that combines the computational reliability of SymPy with the conversational intelligence of a locally hosted LLM through Ollama.
- NeuroCalc addresses this need by integrating both technologies into a single, privacy-focused platform for accurate computation and intelligent mathematical assistance.

Literature Review:

#	Author(s)	Year	Title	Methodology	Contribution	Limitation
1	Nilesh Deotale, Ajay Sirsat, Qniza Shaikh, Haamid Raza Kazi	2026	**Visual Math Interpreter using OpenCV and Generative AI**	OpenCV-based image processing combined with generative AI to interpret visual/handwritten math.	Enables interpretation of handwritten/visual mathematical expressions by combining computer vision with generative AI.	Interpretation-only; limited symbolic verification of computed results.
2	T. D. B. S. Ruberu, Samantha Rajapaksha	2026	**Implementation of a Web-Based Complex Formula Editor**	Web-based editor for authoring and rendering complex mathematical formulas.	Provides a web-based platform for authoring and rendering complex mathematical formulas.	An editing tool only; no AI reasoning, natural-language input, or automatic solving.
3	Sen Shen, Ning Liu, Lintao Song, Dongkun Han	2026	**MathScrub: Single-Pipeline Strikeout Removal and LaTeX Generation for Handwritten Calculus Expressions**	Single pipeline that removes strikeouts and generates LaTeX from handwritten calculus expressions.	Introduces a unified pipeline for strikeout removal and LaTeX generation from handwritten calculus work.	Preprocessing/LaTeX conversion only; does not generate solutions or explanations.
4	Ntsako Morotha Mahlaola, Mogamat Yaseen Martin	2026	**Developing a Proof-of-Concept Automarker for Handwritten Mathematics Assessments**	OCR-based automarker prototype for grading handwritten mathematics assessments.	Develops an OCR-based prototype for automatically grading handwritten mathematics assessments.	Limited to answer verification/grading; no interactive tutoring or reasoning support.
5	Salna Joy, Bipin Sai Surya Guttula, Aditya Gopinath, Bhupalam V. Mohit, Aryan Mishra	2025	**An Intelligent Virtual Canvas for Real-Time Handwritten Equation Recognition and Solution**	Real-time virtual canvas that recognises handwritten equations and computes their solutions.	Delivers a real-time virtual canvas that recognizes and computes solutions for handwritten equations.	Supports limited equation types; minimal conceptual or step-by-step explanation.

#	Author(s)	Year	Title	Methodology	Contribution	Limitation
6	Prathiksha T. D., Preetham Raj Gowda, Mohammed Awez, Nithish Gowda Y. A., Nithin Kumar	2025	**Air Writing Recognition System for Real-Time Mathematical Expression Interpretation**	Tracks air-writing gestures to interpret mathematical expressions in real time.	Introduces a novel air-writing input modality for real-time interpretation of mathematical expressions.	Novel input modality only; lacks a computation and verification backend.
7	Reevan Coelho, Veerasagar S. S., M. S. Srividya, Vinay V. Hegde	2025	**La-Math-ex: LaTeX Code Generation for Handwritten Mathematical Expressions**	Deep-learning model that generates LaTeX code directly from handwritten expressions.	Develops a deep-learning model that directly generates LaTeX code from handwritten mathematical expressions.	Conversion-only; no reasoning, computation, or explanation layer.
8	Sharayu Lokhande, Suraj Jagadale, Suraj Kumar, Yuvraj Padhy, Setlem Lokesh, Shilpa Pawar	2025	**Converting Images of Mathematical Equation into LaTeX Equivalent**	Image-to-LaTeX conversion pipeline for handwritten or printed equations.	Builds an image-to-LaTeX conversion pipeline that handles both handwritten and printed equations.	Conversion-only; provides no computation or conceptual explanation.
9	Leonard U. Ambata et al.	2025	**A Machine Learning-Based OCR System for LaTeX Conversion of Mathematical Expressions**	ML-based OCR system that converts mathematical expressions to LaTeX.	Develops an ML-based OCR system for converting mathematical expressions into LaTeX format.	OCR/conversion focus; lacks AI reasoning for solving or verification.
10	Dhanush Saravanan, Murugesh Muthuraj, Prasanth Rangasamy, S. Raja, D. Tamizharasan	2025	**AI-Powered Real-Time Math Equation Editor with Multi-Modal Input for Enhanced Interaction**	Multi-modal (text, handwriting, voice) equation editor with AI-assisted input.	Creates an AI-powered, multi-modal (text, handwriting, voice) equation editor for enhanced interaction.	Editing/interaction focus; limited depth of symbolic computation or verification.

NEUROCALC SYSTEM WORKFLOW

Module-Level Workflow — Intelligent Mathematical Problem Solving System

1. User Input Module

User enters a mathematical problem (equation, expression, or word problem) through web/mobile/app interface.

2. Input Processing Module

Validates input and performs natural language processing (NLP) to understand the problem and extract key information.

3. Mathematical Parsing Module

Converts the input into a structured mathematical representation (e.g., expression tree or LaTeX format).

4. Problem Classification Module

Identifies the type of problem (e.g., algebra, calculus, matrix, statistics, geometry) to choose the appropriate solving approach.

5. Solver Selection Module

Selects the most suitable solver or engine (symbolic, numerical, or AI-based) based on the problem type.

6. Solving Engine Module

Performs mathematical computations using the selected engine and generates the solution.

7. Solution Generation Module

Generates step-by-step explanation along with the final answer in a human-readable format.

8. Visualization Module (Optional)

Creates graphs, charts, tables, or diagrams to help users better understand the solution.

9. Storage Module

Stores user queries, solutions, and history for future reference and model improvement.

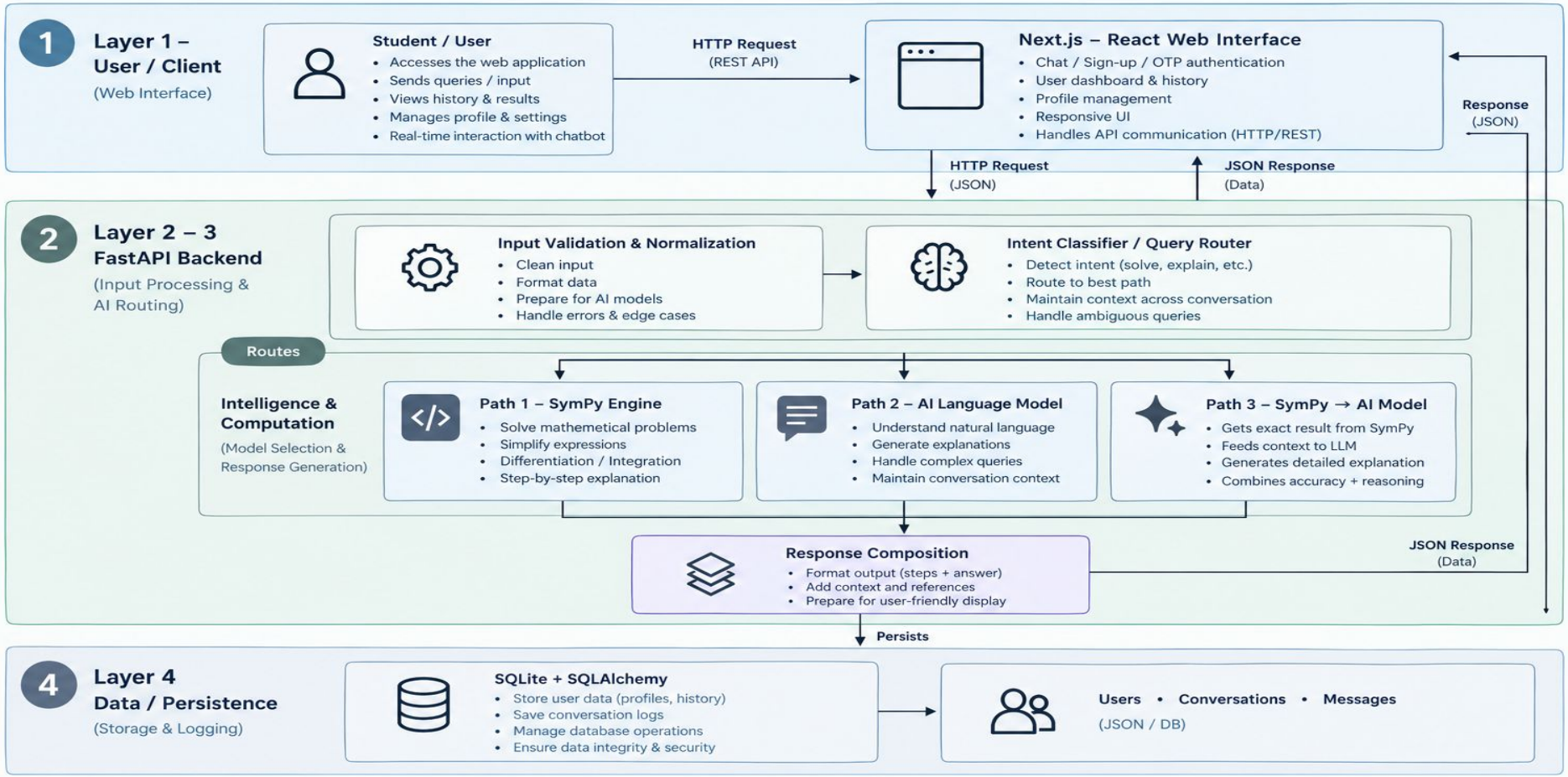
10. Result Delivery Module

Delivers the final result, explanations, and visualizations to the user through the interface.

11. Feedback & Learning Module

Collects user feedback and uses it to improve system performance and accuracy over time.

ARCHITECTURE DIAGRAM



TECH STACK

- Frontend: Next.js, React.js, TypeScript, HTML, CSS
- Backend: Python, FastAPI, Uvicorn
- AI/LLM: Ollama, Llama 3.2 (3B)
- Mathematics: SymPy
- Database: SQLite, SQLAlchemy
- Authentication: Email OTP, Password Hashing
- Email Service: Resend API
- Tools: VS Code, Git, GitHub, Node.js, npm

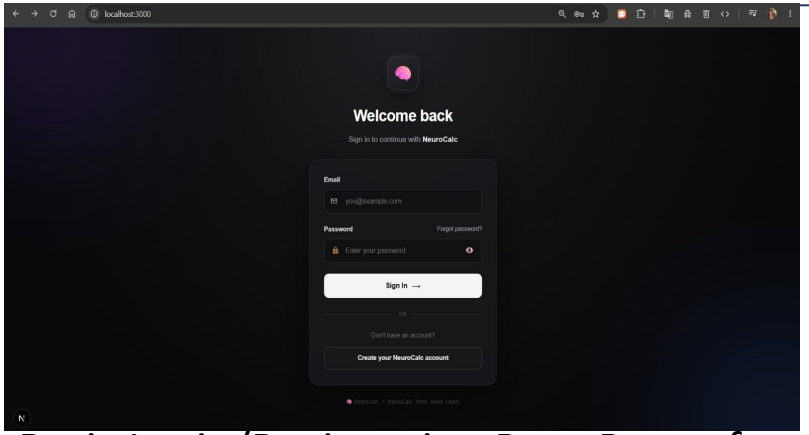
METHODOLOGY

1. **User Authentication:** Secure user registration and login with email OTP verification.
2. **Query Input:** Users enter mathematical problems or learning-related questions through the web interface.
3. **Query Processing:** FastAPI receives and processes user queries.
4. **Intelligent Processing:** SymPy performs mathematical computations, while Ollama with Llama 3.2 generates explanations.
5. **Response Generation:** The system returns solutions and natural-language explanations to the user.
6. **Data Storage:** SQLite stores user details, conversations, and messages.
7. **Learning Support:** Users can review previous conversations to support continued learning.

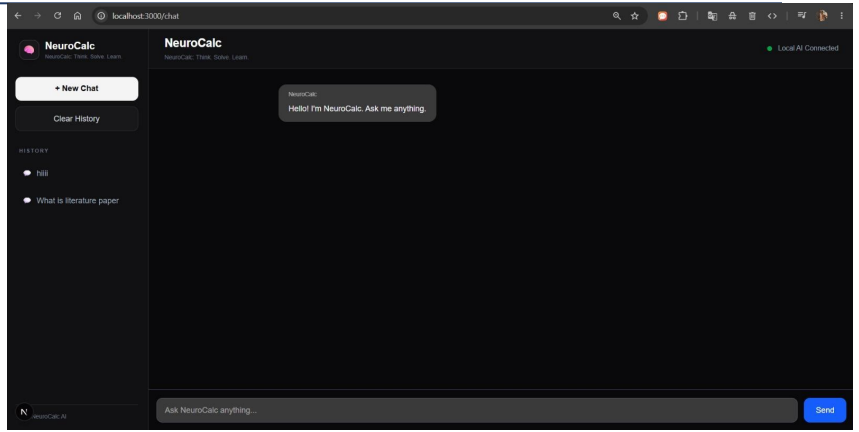
LIMITATIONS

- Accuracy: May produce incorrect answers for complex mathematical problems.
- Hardware: Requires sufficient RAM and processing power to run the local AI model.
- Complexity: May struggle with advanced equations and lengthy reasoning.
- Personalization: Limited adaptation to individual learning needs.
- Internet Dependency: OTP email verification requires an internet connection.
- Scalability: Supporting multiple users simultaneously may require additional resources.

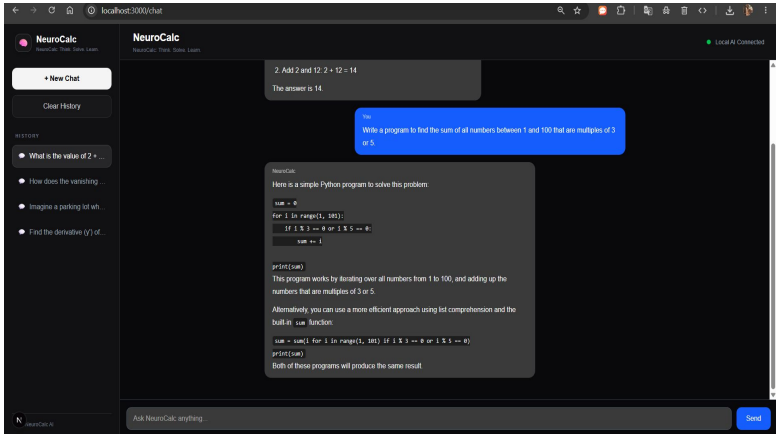
PROJECT IMPLEMENTATION



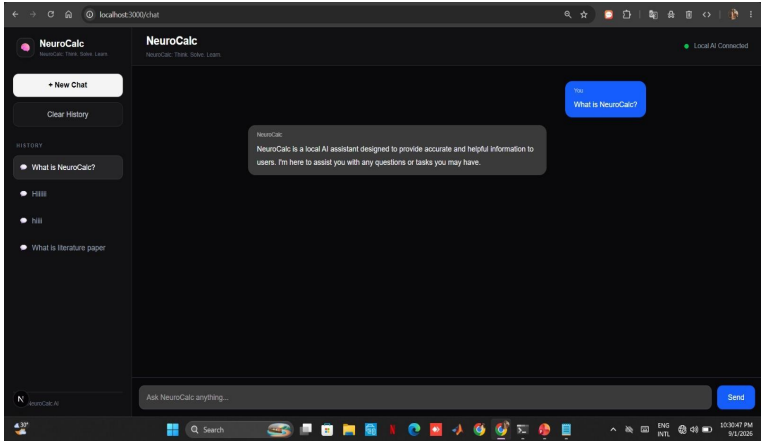
Basic Login/Registration Page Page of NeuroCalc



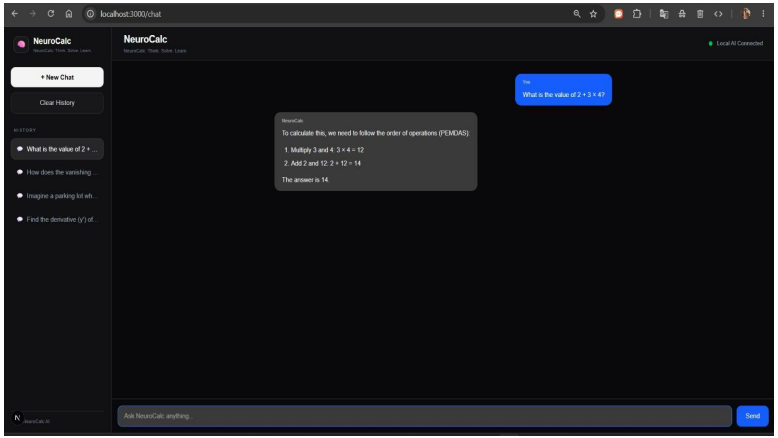
Dashboard of NeuroCalc



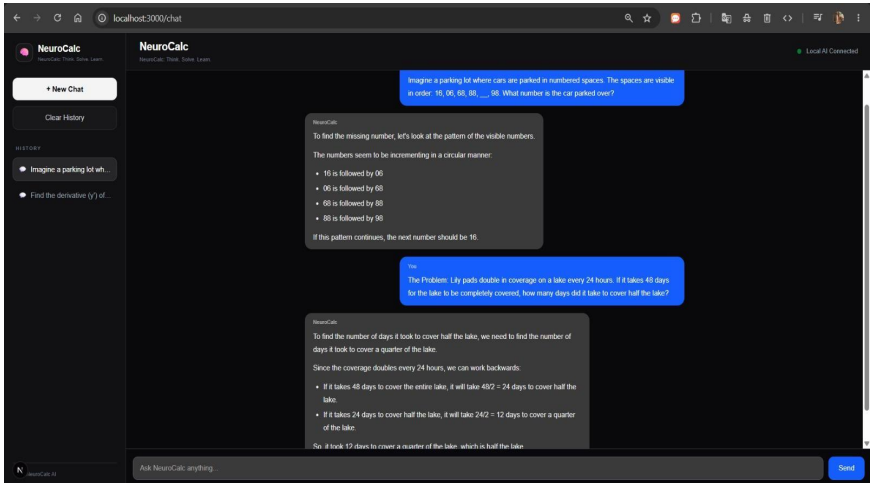
Answering programming-based questions



Answering Definition-Based Question



Answering Mathematical questions



Answering Logical reasoning questions

PROJECT GANTT CHART

Indicative 14-week project schedule

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
1. Problem identification & planning	■	■												
2. Literature survey	■	■	■											
3. Problem statement & objectives		■	■											
4. Research gap analysis			■	■										
5. System workflow & architecture				■	■									
6. NLP & query processing					■	■	■							
7. Mathematical engine						■	■	■						
8. AI explanation generation							■	■	■					
9. Error detection & verification								■	■	■				
10. User interface development									■	■	■			
11. System integration & testing										■	■	■		
12. Evaluation & documentation											■	■	■	
13. Final PPT & demonstration													■	■

Milestone: Working prototype with NLP, mathematical computation, AI explanations, error verification, user interface, integration/testing, evaluation, documentation and final demonstration.

THANK YOU

Neurocalc: An AI-Powered Mathematical Problem-Solving Assistant

FIANL REVIEW PRESENTATION